

Define image segmentation and write its uses. Explain line and edge detections. Write the basic algorithms of Canny edge detector.

image segmentation
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Image Segmentation:

Image segmentation is a computer vision technique that partitions an image into multiple segments or regions to identify objects and boundaries, enabling detailed image analysis and processing. It classifies individual pixels and groups them into meaningful segments based on features like color, brightness, and texture.

Uses:

Medical Imaging: Detect and segment organs, tumors, or anomalies in medical scans.

Self-Driving Cars: Identify and track roads, vehicles, and pedestrians.

Robotics: Understand environment by segmenting objects.

Satellite Imagery: Identify and map land use or specific objects.

Image Editing: Separate objects from backgrounds .

Explain line and edge detections.

Line Detection

Second Derivatives: Yield thinner lines and stronger responses than first derivatives, though they require handling of the **double-line effect**.

Directional Masks: Use specific kernels for horizontal, vertical, and ± 45 degree orientations.

Detection Logic: Let R_1, R_2, R_3, R_4 be the mask responses. A point is associated with direction k if:

$$|R_k| > |R_j| \text{ for all } j \neq k$$

Edge detection

is a technique in image processing that is used to find the boundaries of objects in an image. It works by identifying sudden changes in brightness or color, which usually indicate where one object ends and another begins. Common methods for edge detection include Sobel and Canny, with Canny being the most accurate and widely used.

Canny Edge Detector Algorithm

1. **Gaussian Smoothing:** Reduces image noise using a Gaussian filter.
2. **Gradient Calculation:** Finds magnitude $M(x, y) = \sqrt{g_x^2 + g_y^2}$ and direction $\alpha(x, y) = \arctan(g_y / g_x)$.
3. **Non-maximum Suppression:** Thins edges by retaining only local maxima along gradient directions.
4. **Hysteresis Thresholding:** Uses dual thresholds (T_H, T_L) to link strong edges and connected weak pixels while discarding isolated noise.